ROYE FANG

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Education

University of California, Los Angeles [Expected Graduation – Spring 2023]
B.S. Computer Engineering
Cumulative GPA – 3.528

Technical Skills

Programming Languages
- C/C++, Python

Software
- Linux, MATLAB

Electrical Engineering Applications
- Arduino, SMD Soldering, EAGLE (PCB Design), I2C/SPI/UART Protocol

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Practical Experience

Institute of Electrical and Electronics Engineers (IEEE) [September 2019 – Present]
Open Project Space Team Member [UCLA]
- Breadboarded and soldered circuits using different electronic components and ICs
- Utilize Arduino Nano to build interactive games, such as "Red Light, Green Light" with UART
- Soldered a distance sensor to change LED brightness using a phototransistor and IR emitter
- Designed schematic and PCB for an autonomous maze-navigating car capstone project

MicroMouse Team Member
- Built and powered a robot "mouse" with DC motors to solve an arbitrary 16×16 maze
- Utilized IR sensors and PID feedback loop to implement multiple maze-solving algorithms
- Competed with other teams for the fastest maze-solving time and most consistent navigation

iD Tech Camps [July 2020 – Present]
Online Instructor [Campbell, CA]
- Taught K-12 students in subjects relating to C++, Scratch, Arduino, and Minecraft
- Developed unique programming and technology curriculum for individual private lessons
- Strengthened interpersonal connections with students and clientele in a remote setting

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Projects

passwrd – Secure Password Generator [February 2021]
- Wrote a **Python** script that randomly generates passwords for online user accounts
- Conducted online research to determine the characteristics of strong and secure passwords
- Implemented options to generate arbitrary passwords and hashing phrases into passwords

time.out – Bluetooth-controlled Alarm Clock [January 2021]
- Designed and built a Bluetooth controlled alarm clock for IDEAHacks (hardware hackathon)
- Programmed ESP32 to power a liquid crystal display UI, 7 segment displays, and a buzzer
- Simulated user navigation and option menus through a Serial Bluetooth Terminal application

TI–RSLK – Autonomous Line-following Car [May 2020]
- Applied PWM signals on TI-RSLK to simulate steering and braking using Energia IDE
- Conducted sensor fusion to normalize eight different IR sensors to be used for navigation
- Programmed a PID feedback algorithm to accurately follow the multiple, arbitrary paths
- Scored fastest overall completion time for all tracks/races in a twenty person lab section

Mori – Short-term Memory Game [April 2020]
- Coded a short-term memory game in Arduino by flashing a string of numbers in serial monitor
- Created a user input module using three push buttons to enter the string of numbers
- Designed customizable game settings and scoreboard system over multiple rounds/iterations

iPoduino – Arduino Music Player [January 2020]
- Built an 8-bit music player using Arduino Nano MCU to control a passive buzzer
- Programmed multiple full-length piano pieces and pop songs using "pitches.h" library
- Implemented "shuffling" through multiple songs using a potentiometer and RGB LED